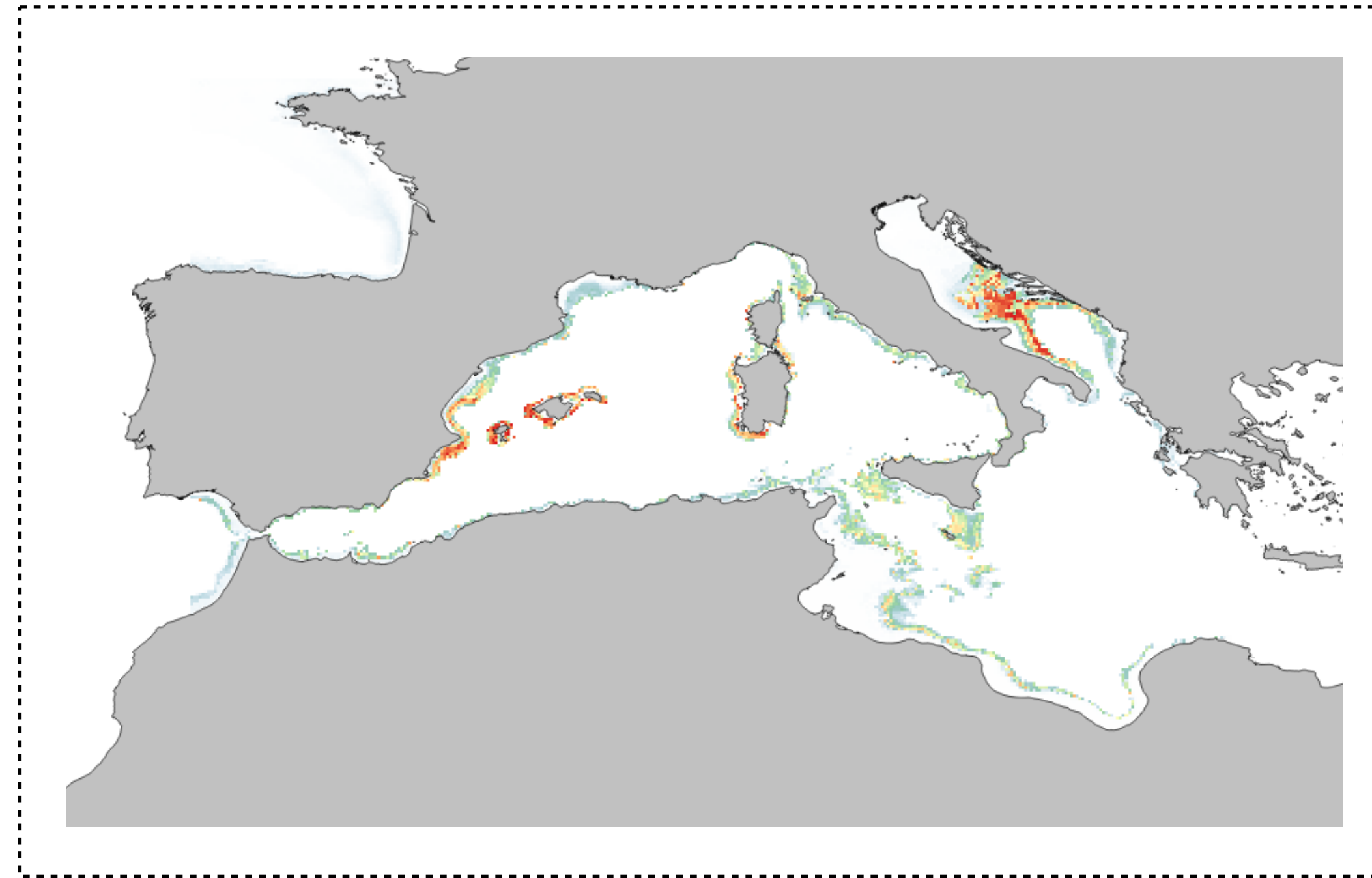


# Evaluating predictive performance

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal





## Model evaluation / performance

### Key goals:

Verify consistency: do predictions match observations (presences and absences)?

Assess transferability: how well does the model generalize to new data?

Evaluate ecological realism: does the model make biological sense?

**Fundamental question:** is the model suitable for the purpose (e.g., impact assessment)?

# Model evaluation

**Prediction errors can be of ‘false positives’**, when the model predicts occurrence in places where the species has not been observed, **and ‘false negatives’**, when the model predicts absent in places where it has been observed.

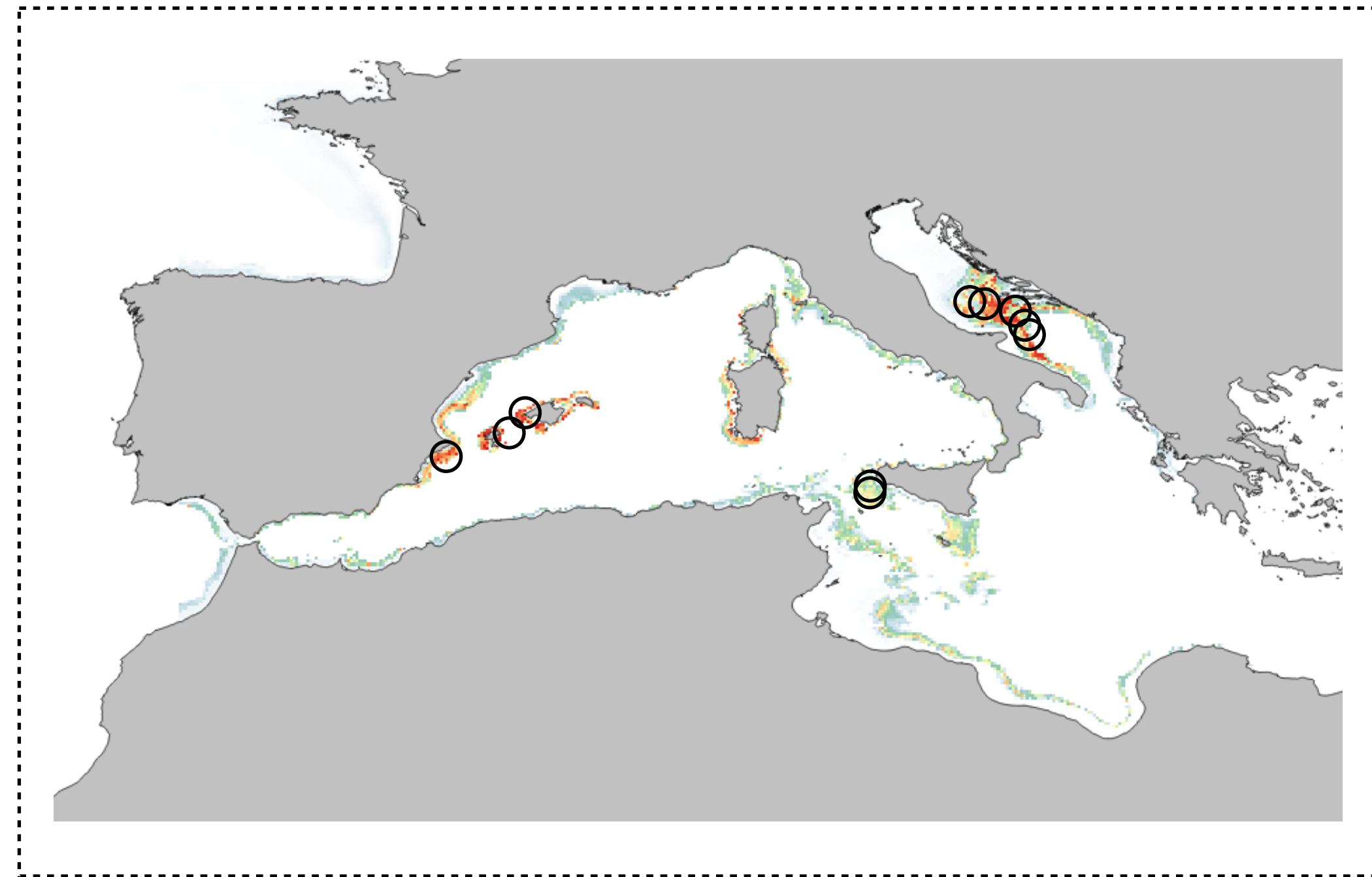
Can be summarized in confusion matrices.

| Confusion matrix<br>Types of prediction errors |          | Observation    |                |
|------------------------------------------------|----------|----------------|----------------|
|                                                |          | Presence       | Absence        |
| Prediction                                     | Presence | True Positive  | False Positive |
|                                                | Absence  | False Negative | True Negative  |

**Perfect models** only retrieve true positives and true negatives.

## Evaluation criteria

Predictions are continuous surfaces (e.g., probability or suitability; from 0 to 1);  
One cannot compare one observation (e.g., 1) with its corresponding model output (e.g.,  $P: 0.7$ ).

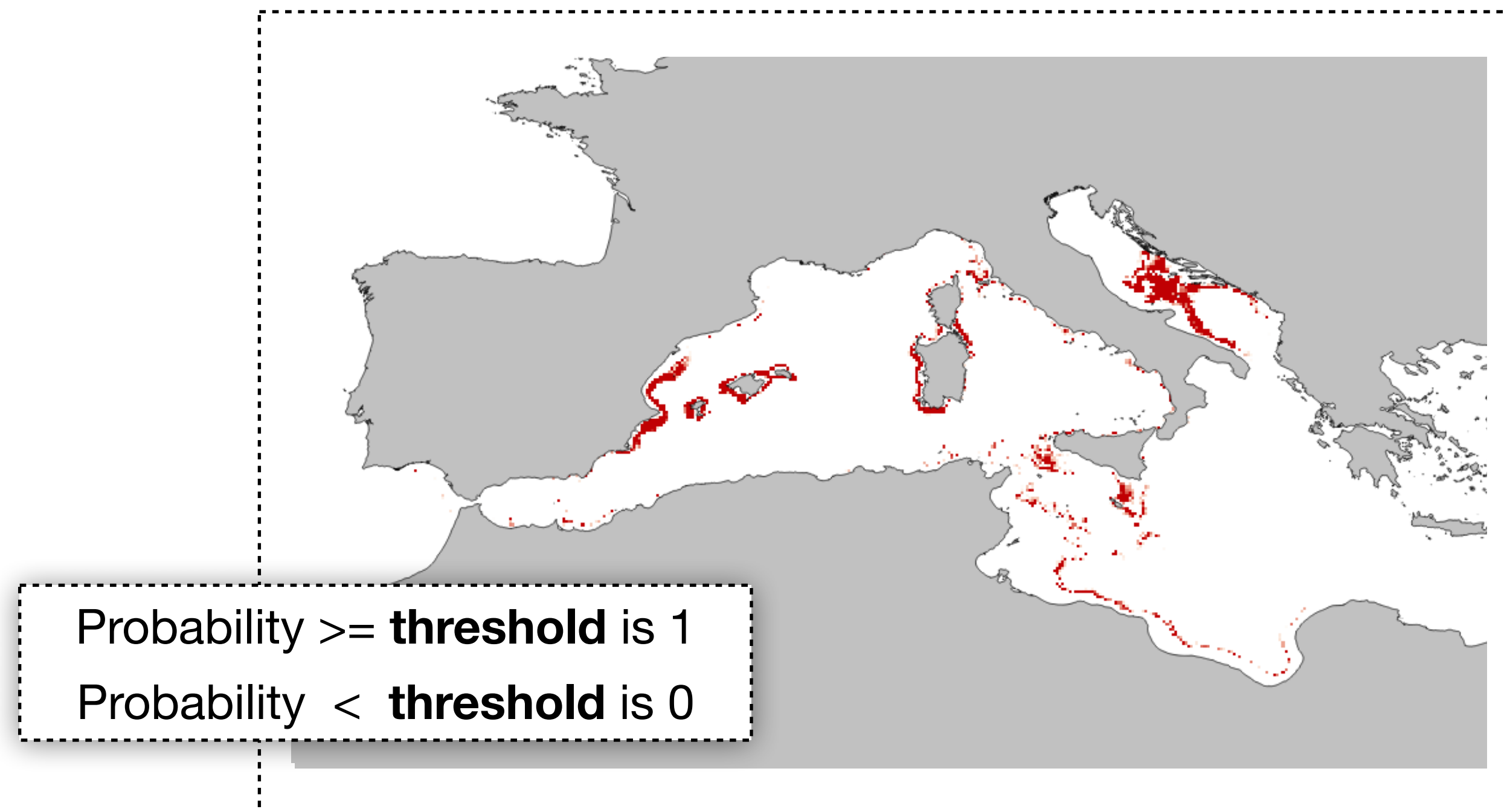


**Predictions need to be reclassified as binary responses** (presence-absence)  
for comparison with the observed data.



## Evaluation criteria

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**Predictions need to be reclassified as binary responses** (presence-absence)  
for comparison with the observed data.

## Evaluation metrics: threshold-dependent

The elements of the contingency table are used to compute various evaluation criteria that measure the performance of the model.

Calculated after applying a threshold to predictions.

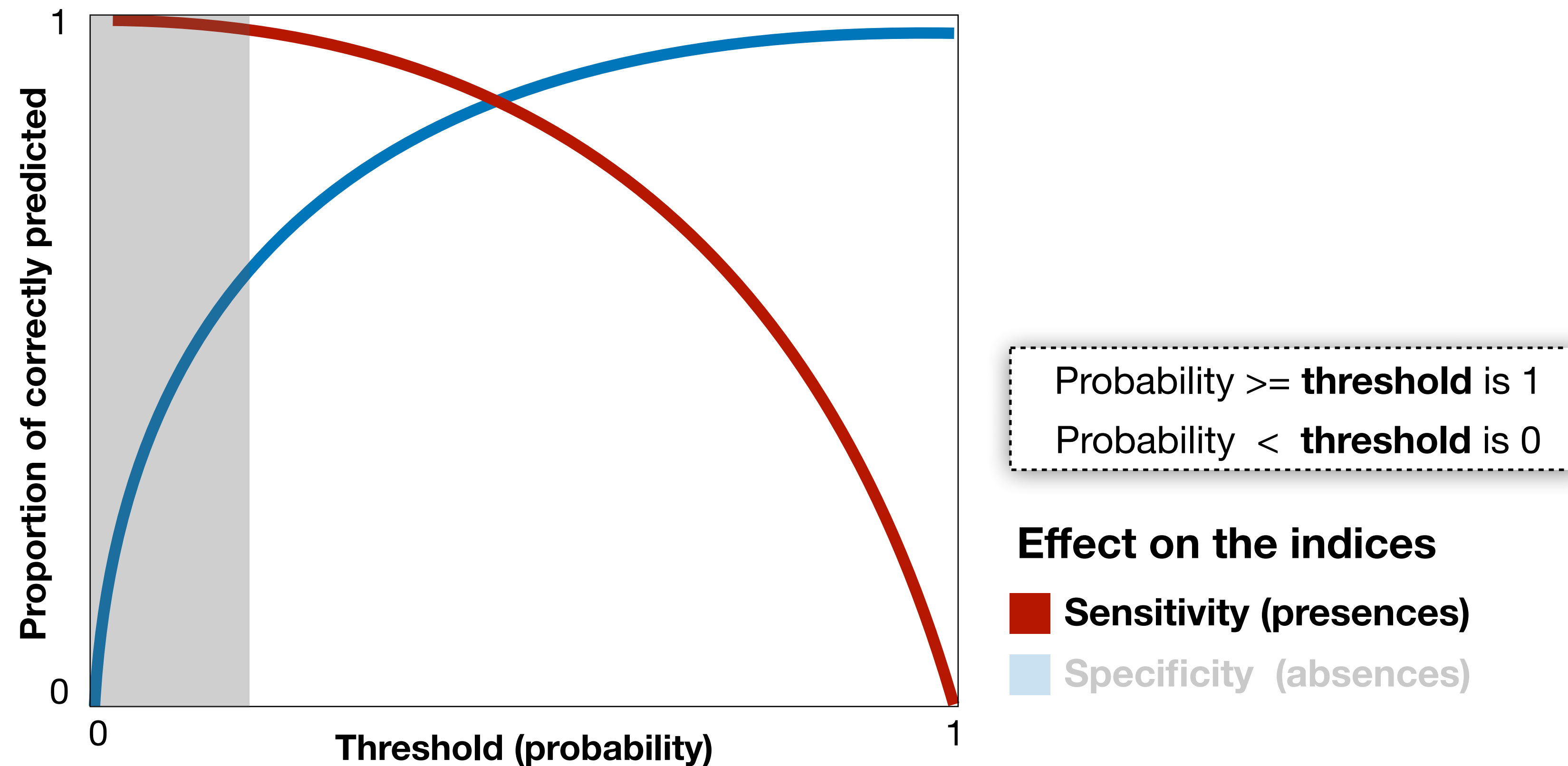
**Sensitivity:** proportion of presences correctly predicted;

**Specificity:** proportion of absences correctly predicted;

**True Skill Statistics:**  $\text{Sensitivity} + \text{Specificity} - 1$  (describes how well the model predicts presences and absences).

## Evaluation criteria

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.

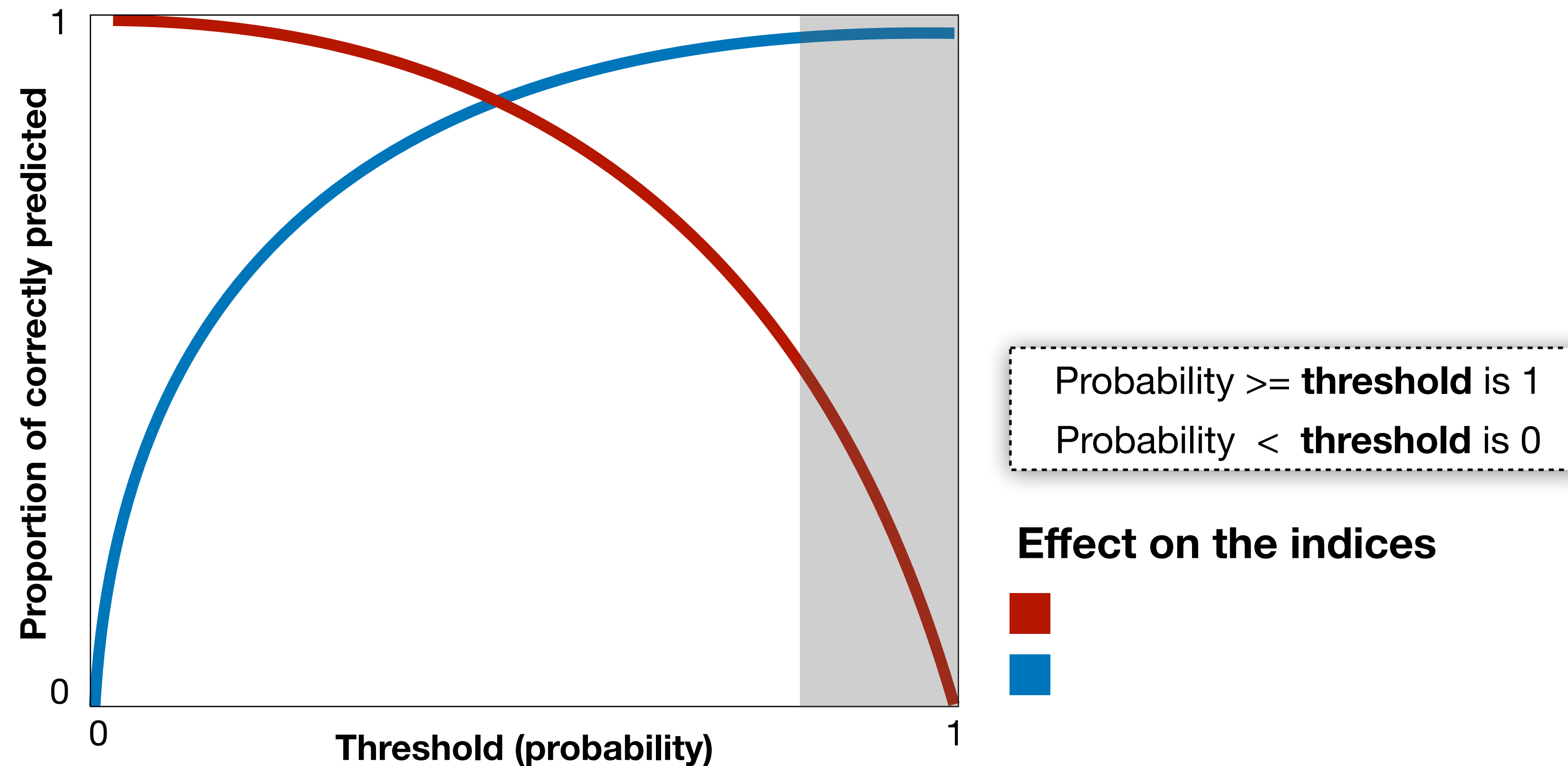


Using a low threshold most cells (in the map) will return 1.



## Evaluation criteria

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.

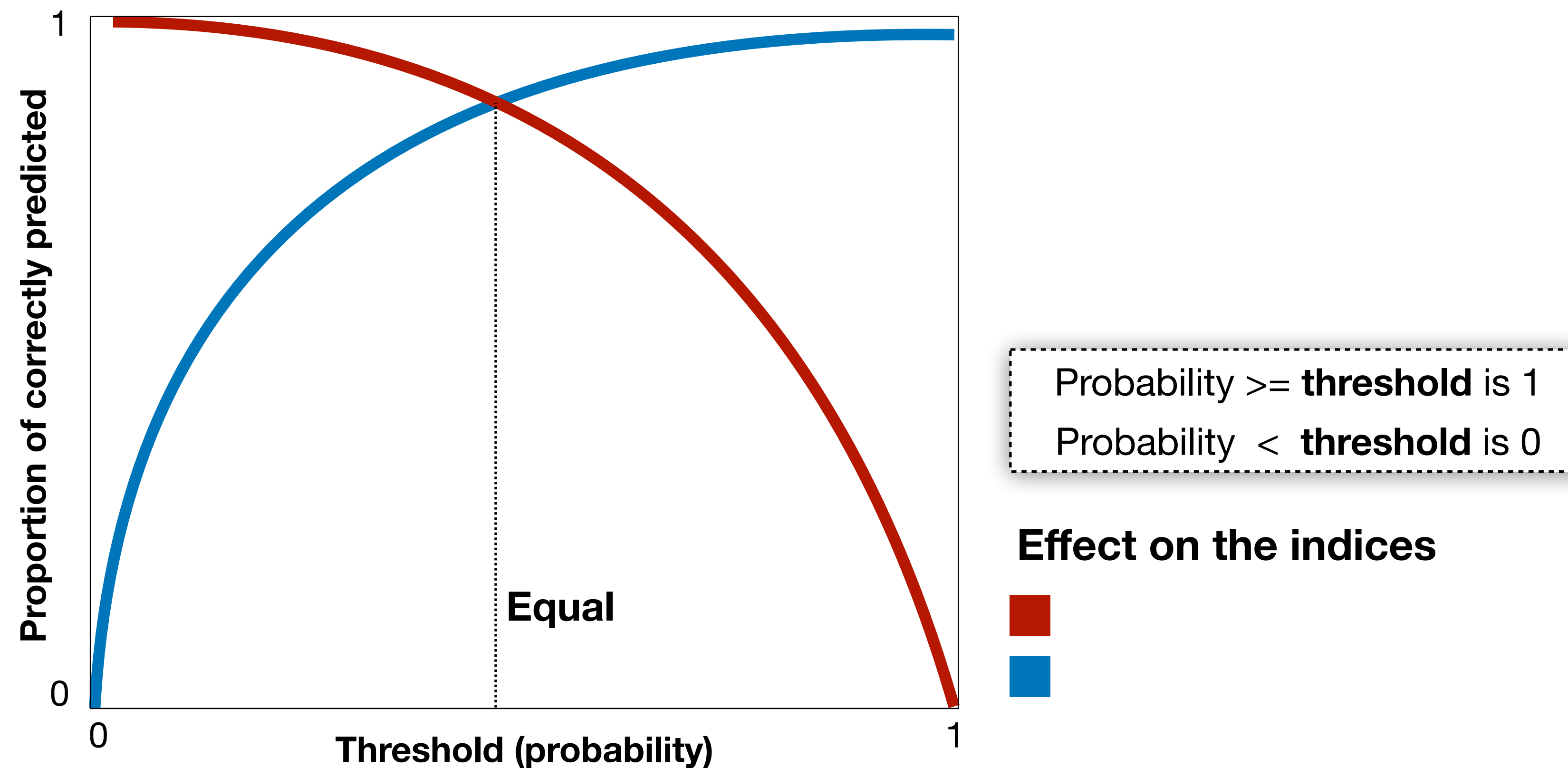


Using a high threshold most cells (in the map) will return 0.



## Evaluation criteria

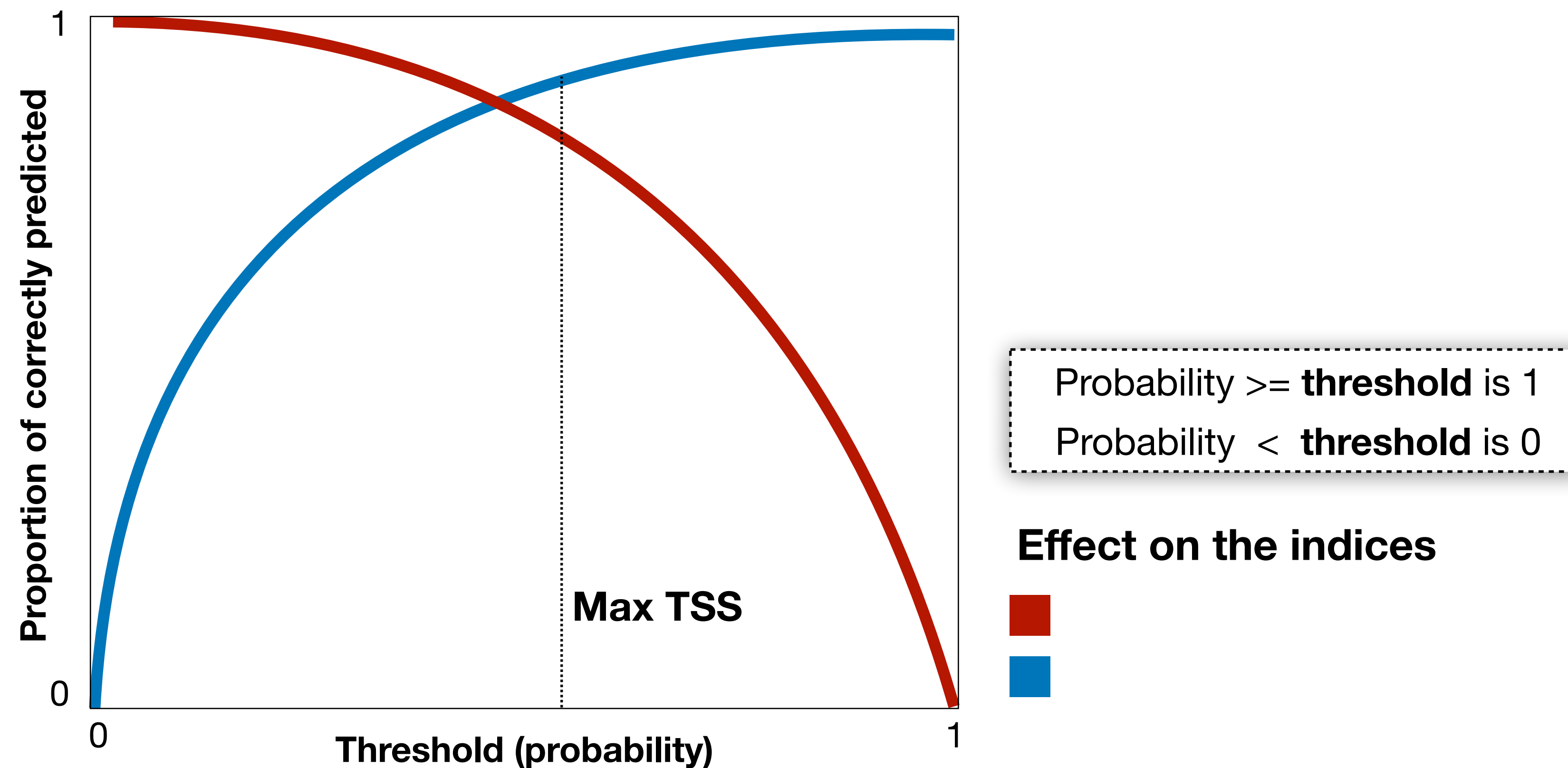
There are threshold rules to maximize the agreement between observed data and the predicted binary surface.



(1) **Equal sensitivity and specificity**

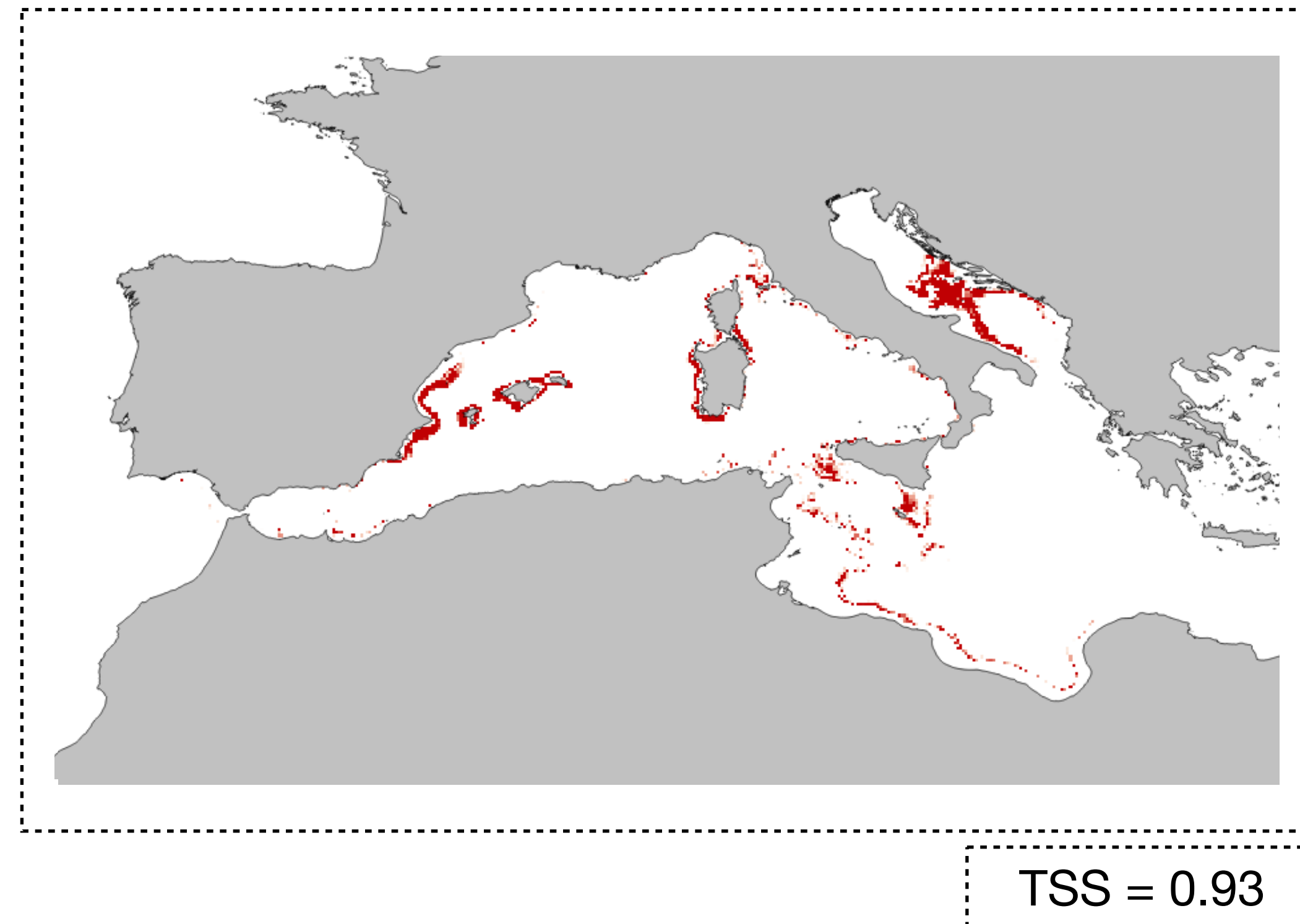
## Evaluation criteria

There are threshold rules to maximize the agreement between observed data and the predicted binary surface.



(2) Maximization of sensitivity + specificity

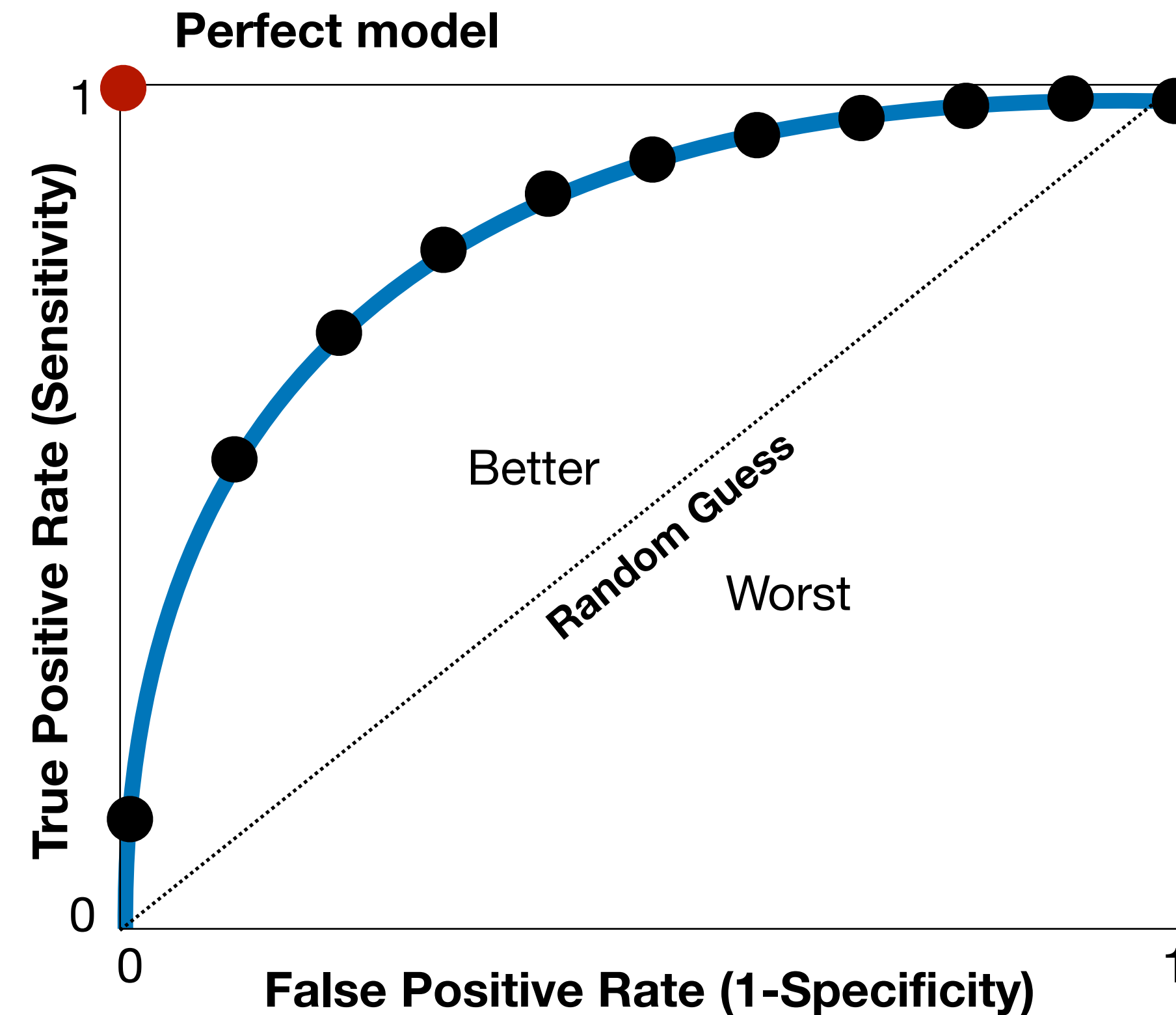




## Threshold-dependent measures of accuracy

Thresholds like “Max. TSS” and “Equal TSS” allow **assessing the agreement between observed data and the model output (e.g., probability of occurrence) and also converting probability maps into binary maps.**

**TSS ranges from -1 to +1 (0 <= no better than a random model).**



## Threshold-independent evaluation criteria

**A**rea **U**nder the **C**urve of the Receiver Operating Characteristic Plot.

A Threshold-independent index contrasting **true positive rate (sensitivity)** against **false positive rate (1-specificity)** across the range of all possible thresholds. The closer the curve from y-axis, the larger the **AUC**, and thus the more accurate the model.

## AUC performance

Accuracy indices like AUC can be interpreted as follow:

- 1 - 0.9 : excellent model
- 0.9 - 0.8 : good model
- 0.8 - 0.7 : fair model
- 0.5 - 0.7 : poor model

### **AUC recommend because:**

A well-known standard in SDM;

### **AUC not recommend because:**

Relies only on presence and absence records (or pseudo-absences / background, which are artificially generated)



## Evaluating SDMs: the Boyce index

It assesses how well the model's predicted suitability values correlate with the distribution of the presence points across the landscape.

### Why is it a good alternative to AUC?

**Designed for presence-only:** Unlike AUC, which requires absences, the Boyce index tests if higher suitability predictions correspond to higher densities of presence points, which is the desired model behavior.

**Boyce index ranges from -1 to +1 (0  $\leq$  no better than a random model).**

## Model evaluation

**Multi-faceted approach:** don't rely on a single metric.

**Use a suite:**

Sensitivity;

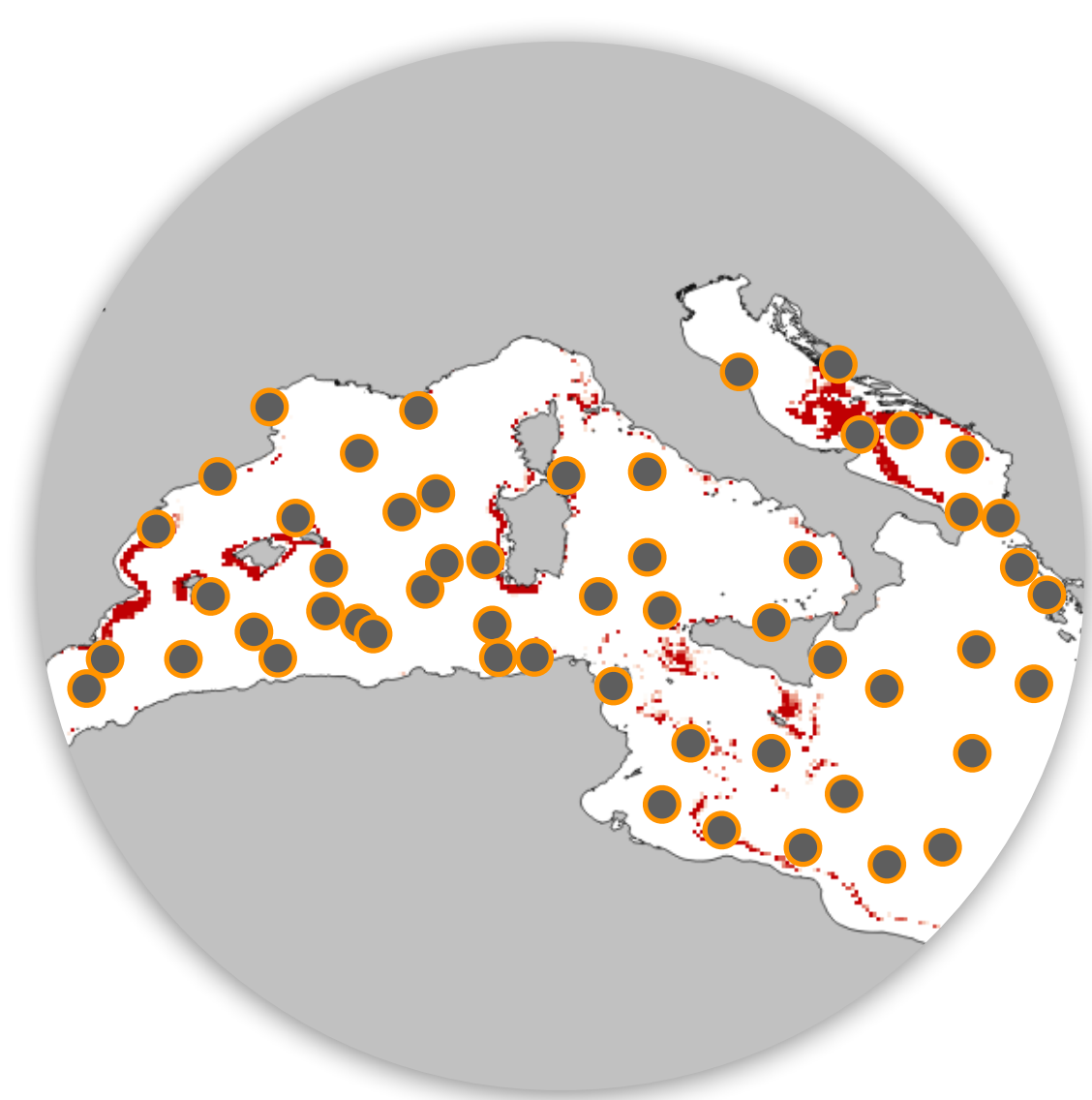
Specificity;

AUC;

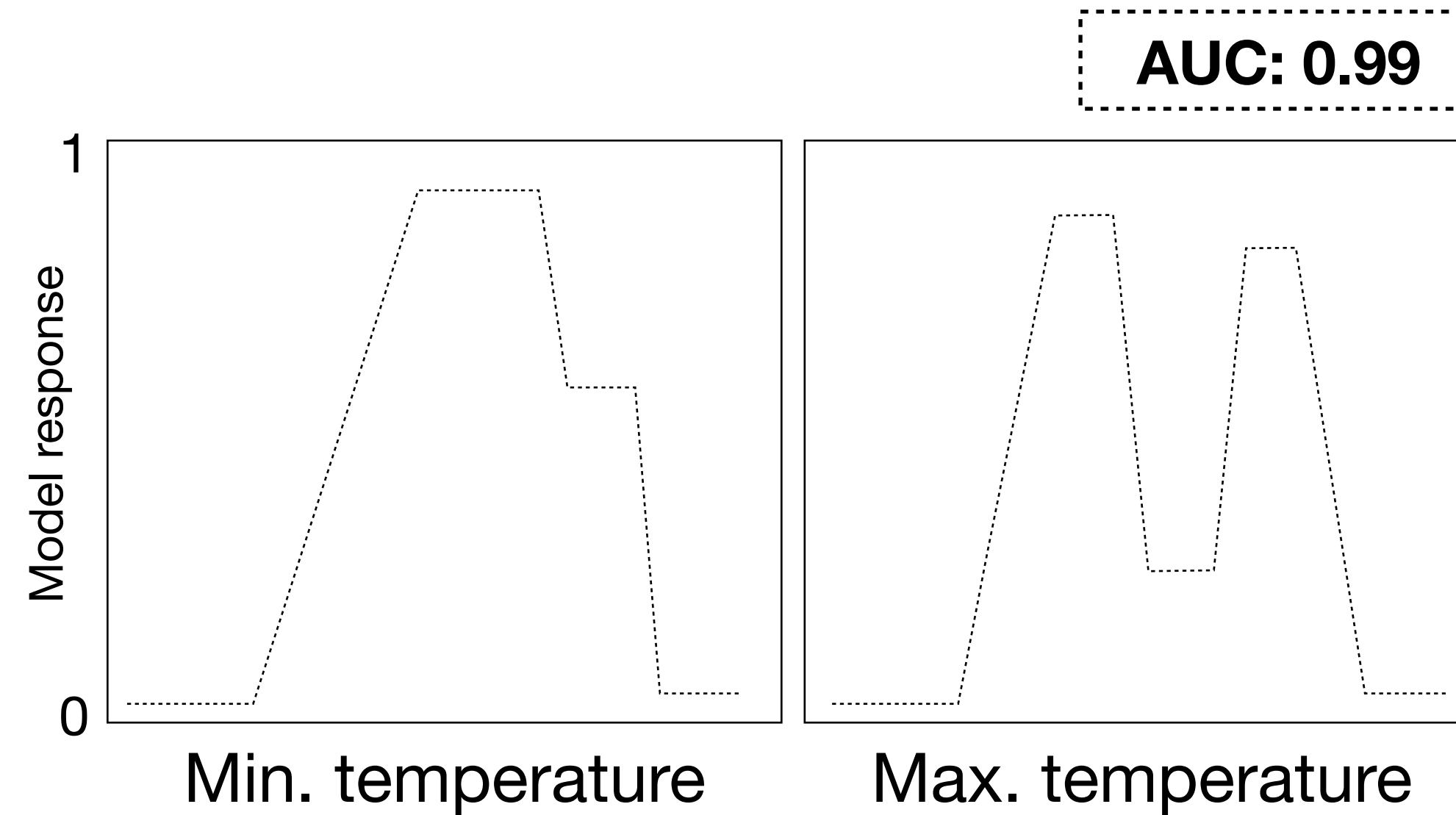
Boyce;

## High performance linked to good model transferability?

Depends on how it is measured.



- Testing data
- Training data

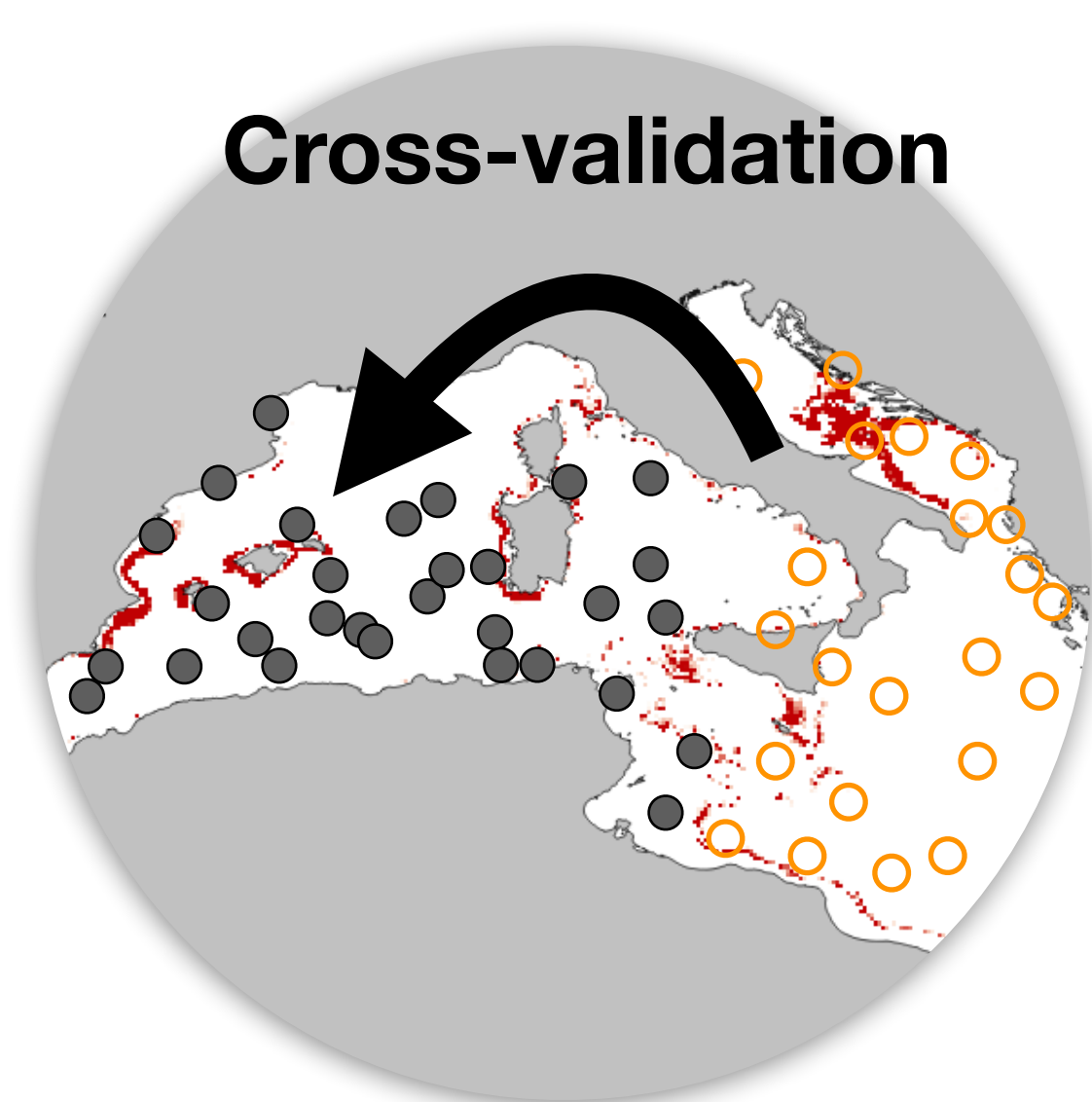


**Testing accuracy with the training data leads to an overestimation of accuracy**, regardless of the model's potential for transferability or if it ecologically sound.

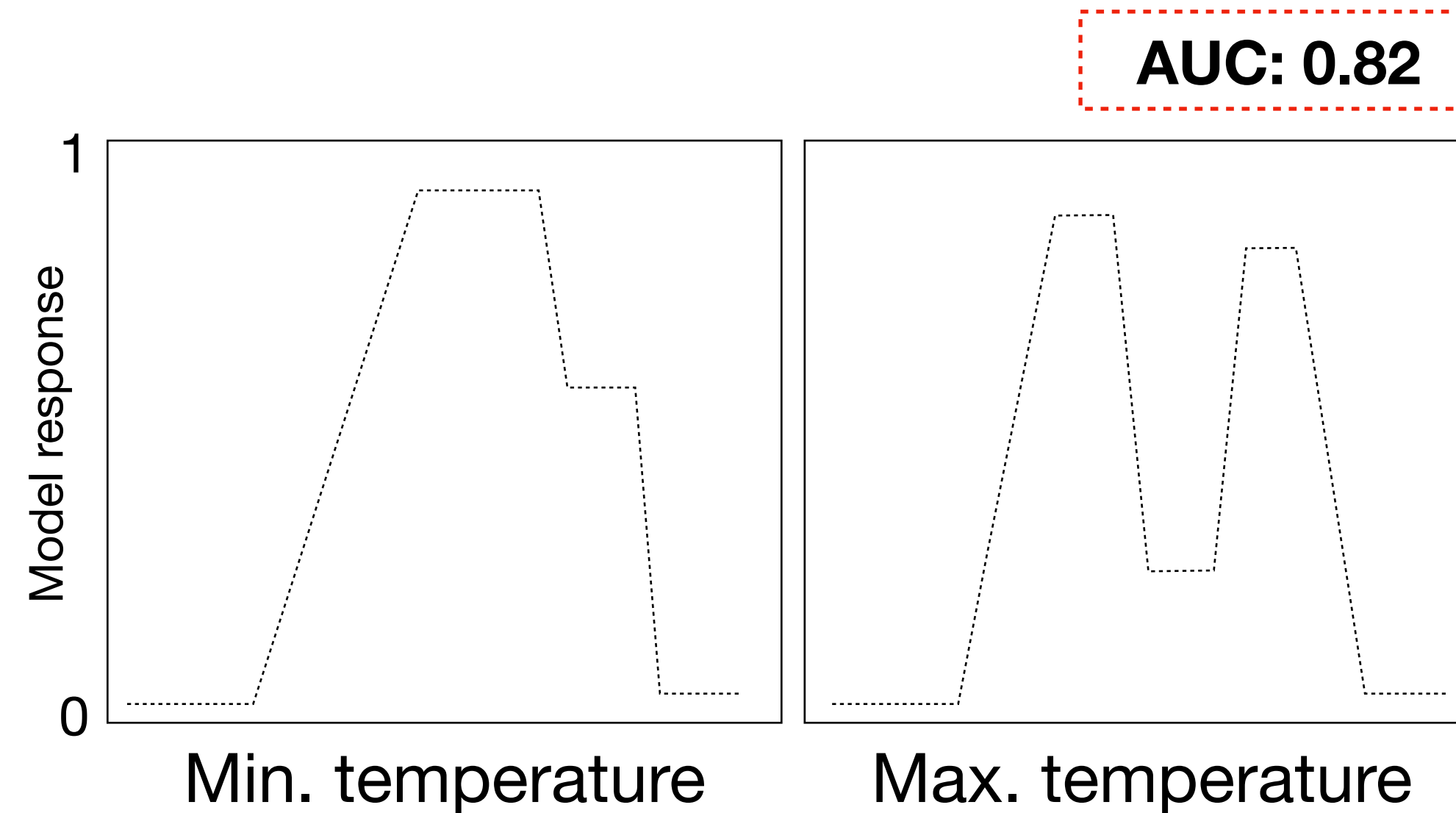


## High performance linked to good model transferability?

Depends on how it is measured.



- Testing data
- Training data



**Testing accuracy with independent data is the approach to evaluate the model and its transferability.**

This is the same as projecting to other places or times.

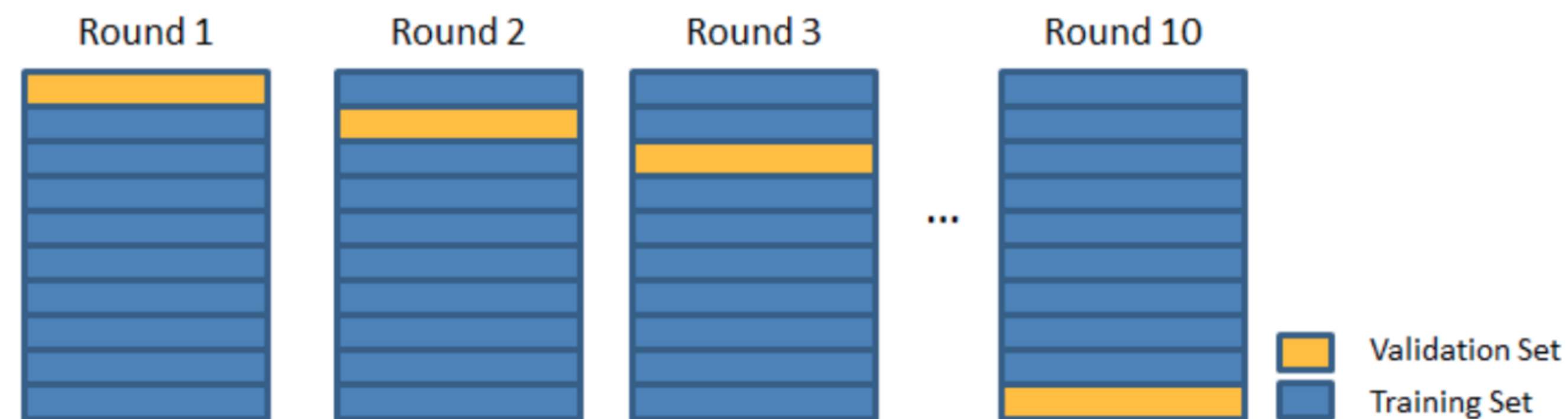
Leads to lower accuracy but to a more reliable score.

## Missing independent data?

**Approach:** partitioning the data in k-fold cross-validation interactions, with data splits k times, yielding k estimates of accuracy that can be averaged.

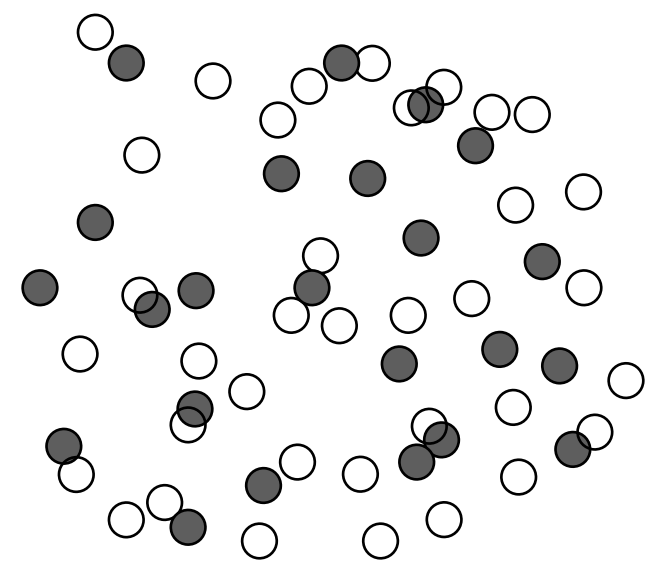
**e.g.,**

In 10-fold CV, 9/10 of the observations are used to train the model and the remaining 1/10 are used to estimate performance; this is repeated ten times and the estimated performance measures are averaged.



## Missing independent data

There are different methods to produce independent datasets.  
Some approaches provide more independent datasets than others.



**Random (70/30)**

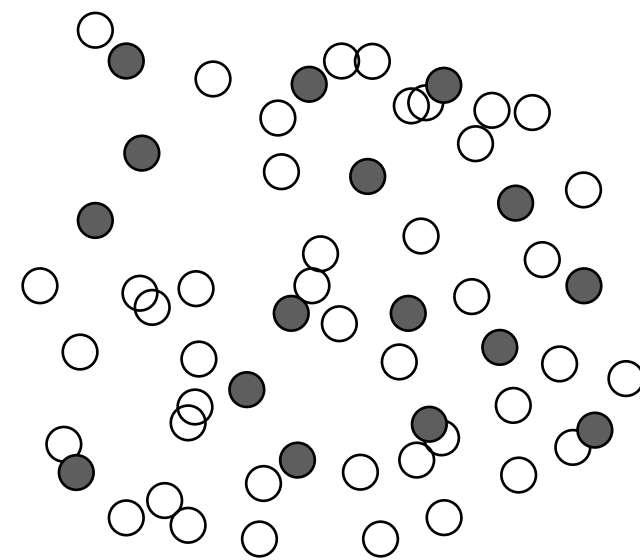
(70/30 | k-fold)

○ Training data    ● Testing data



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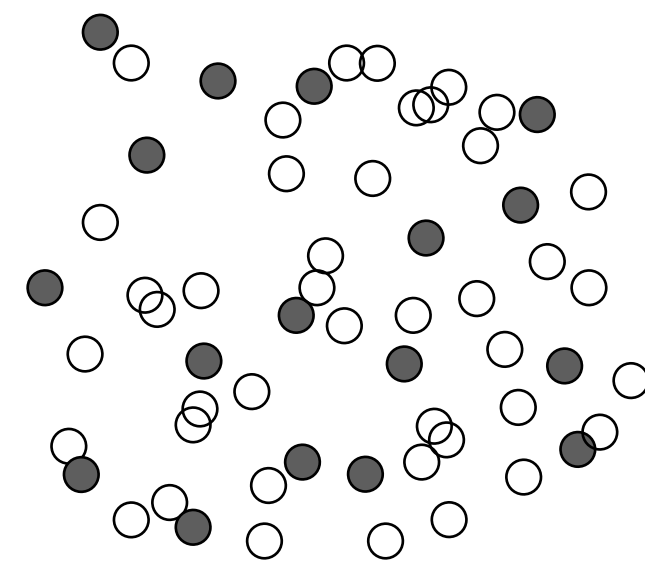
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There are different methods to produce independent datasets.  
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Presence / absence data at nearby locations are often not independent.

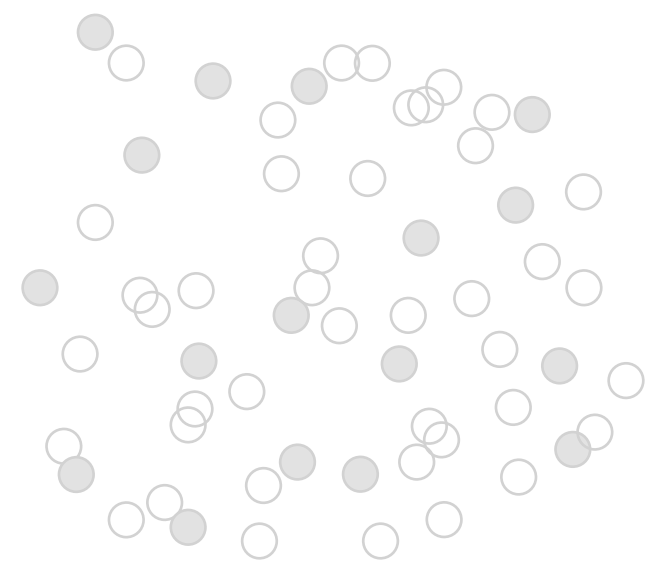
*"Near things are more related than distant things".*

Standard (random) k-fold CV can overestimate performance because training and testing folds might contain similar data.

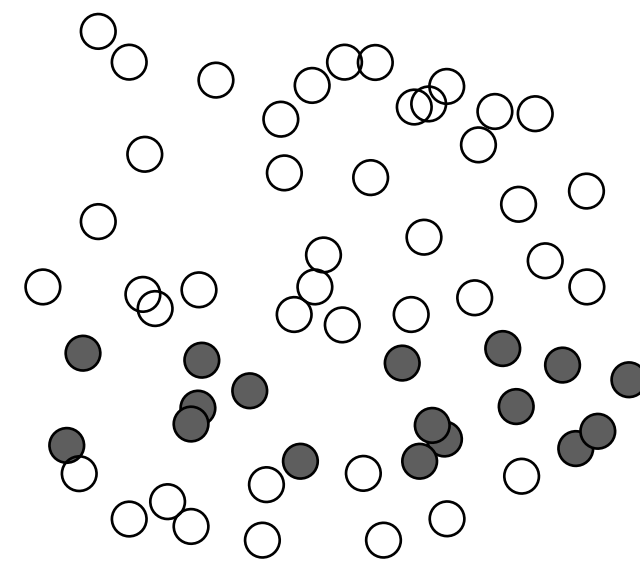
Violates independence assumptions of many statistical tests.

## Missing independent data

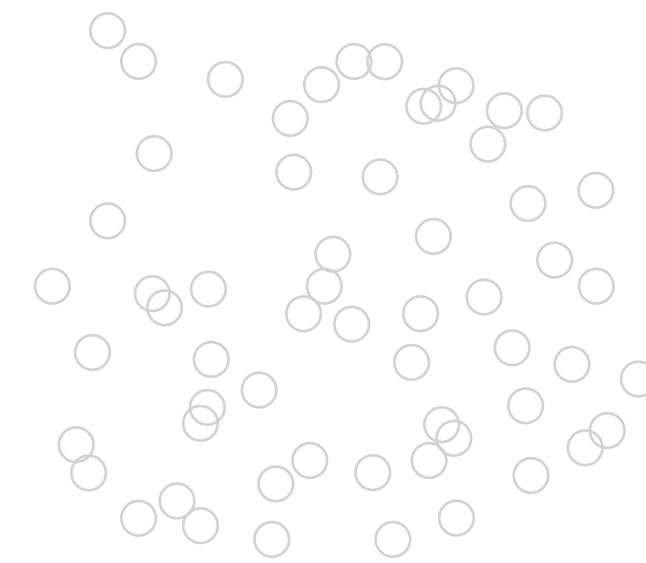
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**Random (70/30)**  
(70/30 | k-fold)



**Bands**  
(latitudinal, longitudinal)



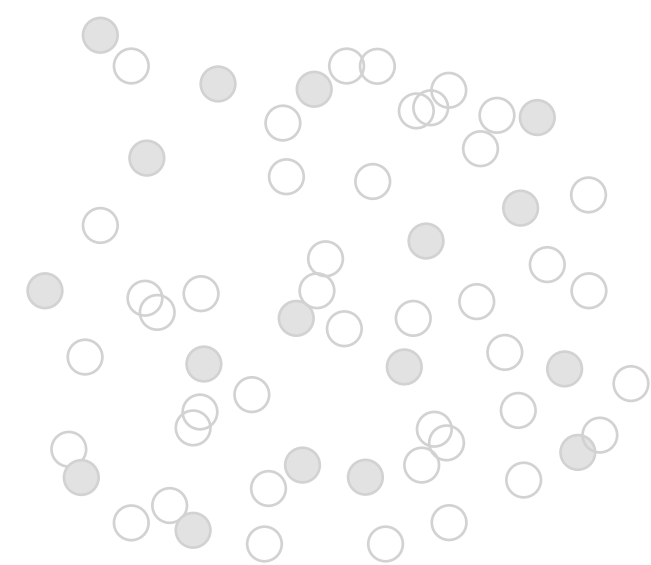
**Blocks**  
(latitudinal, longitudinal)

○ Training data    ● Testing data

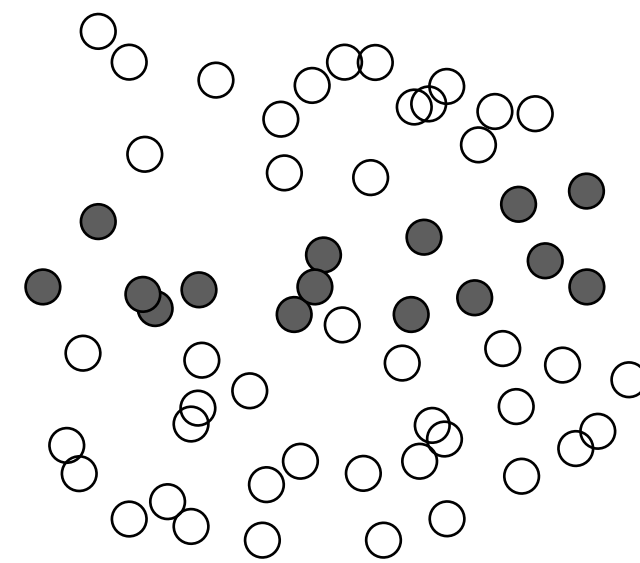


## Missing independent data

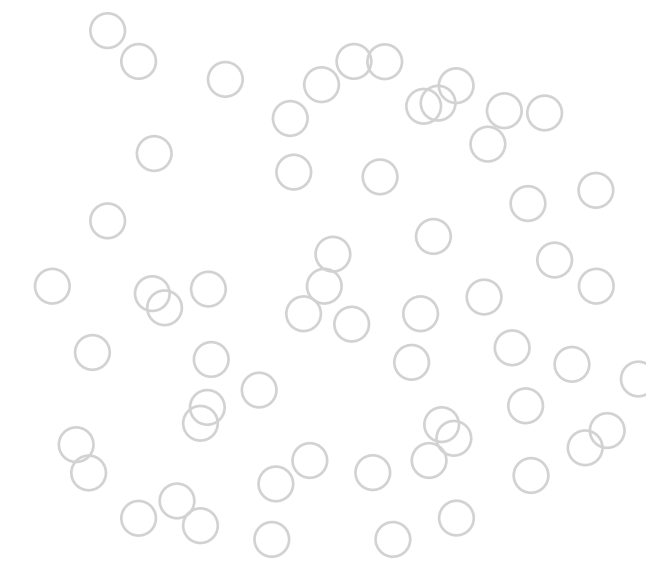
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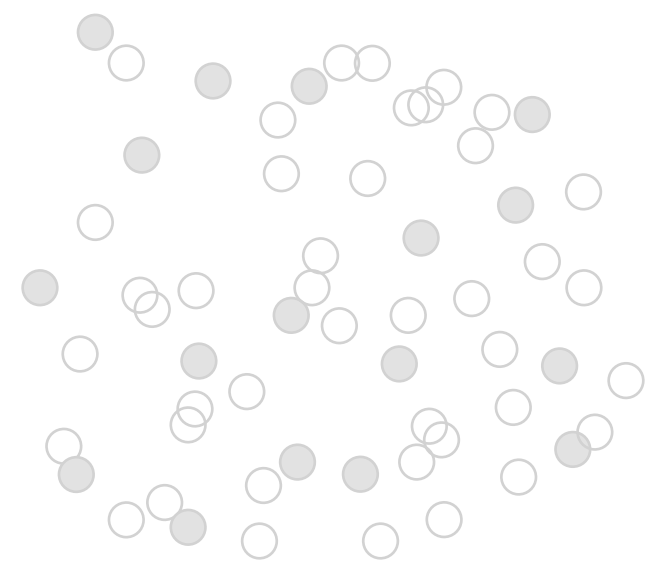
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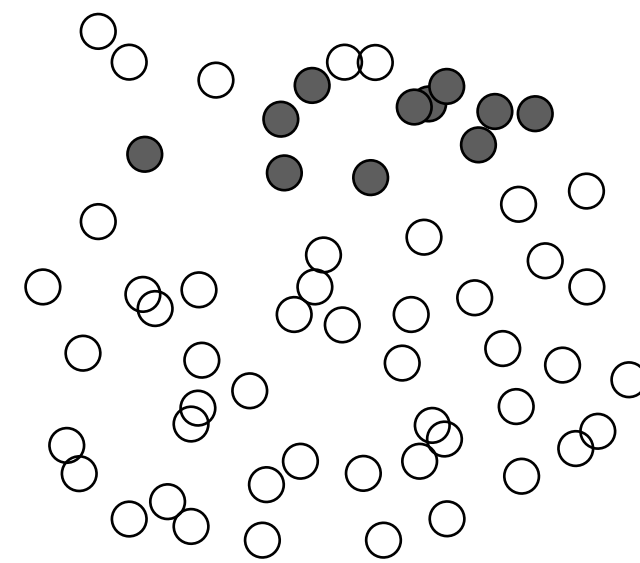


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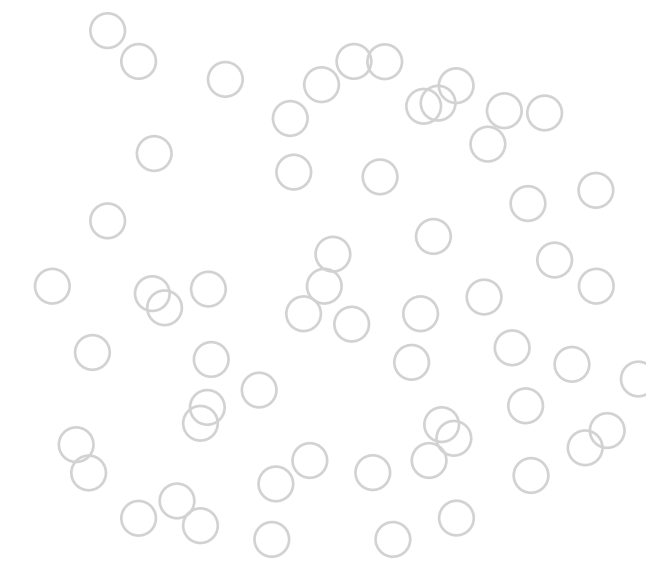
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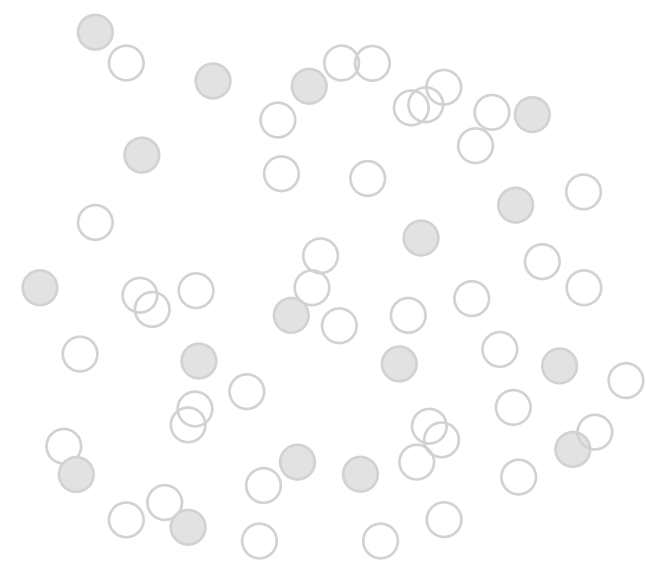


**Blocks**  
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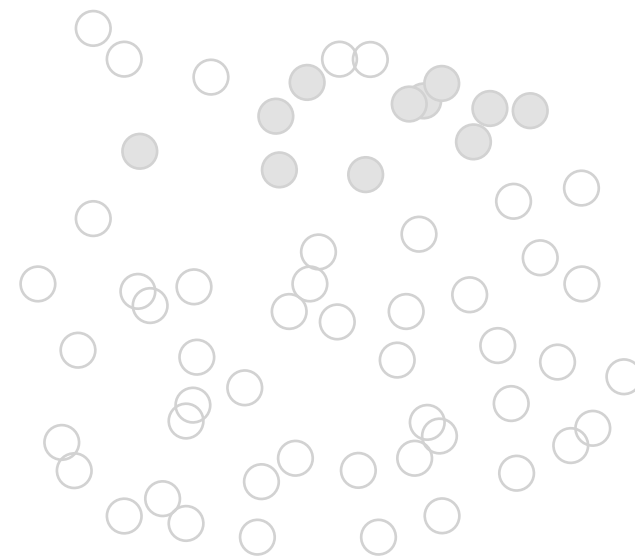
○ Training data    ● Testing data

## Missing independent data

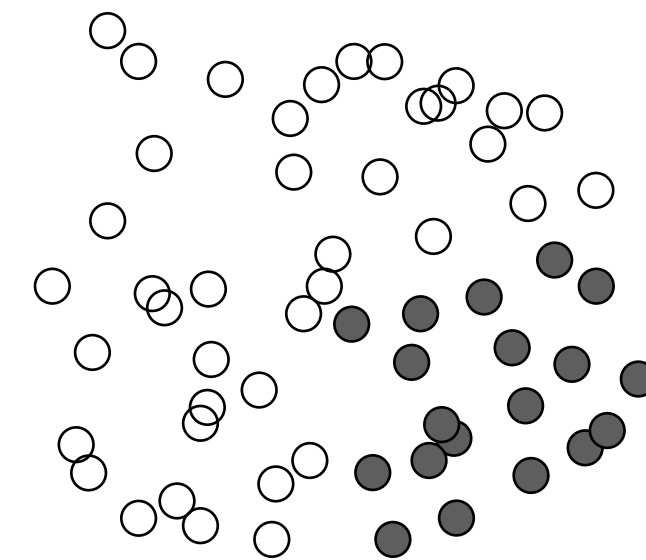
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**Bands**  
(latitudinal, longitudinal)



**Blocks**  
(latitudinal, longitudinal)

○ Training data    ● Testing data

## Performance of algorithms in SDM

Most algorithms have been tested and compared.

**Differences in performance among different algorithm types tend to be smaller than differences among species.**

**Machine learning algorithms**, like MaxEnt, and especially those that incorporate model averaging (e.g., BRT) tend to have **better performance than more simple statistical methods** such as GLMs.



## **Beyond metrics: ecological realism**

**Does the model make biological sense?**

High metric scores (AUC, TSS) are necessary but NOT sufficient!

**Ecological plausibility check:**

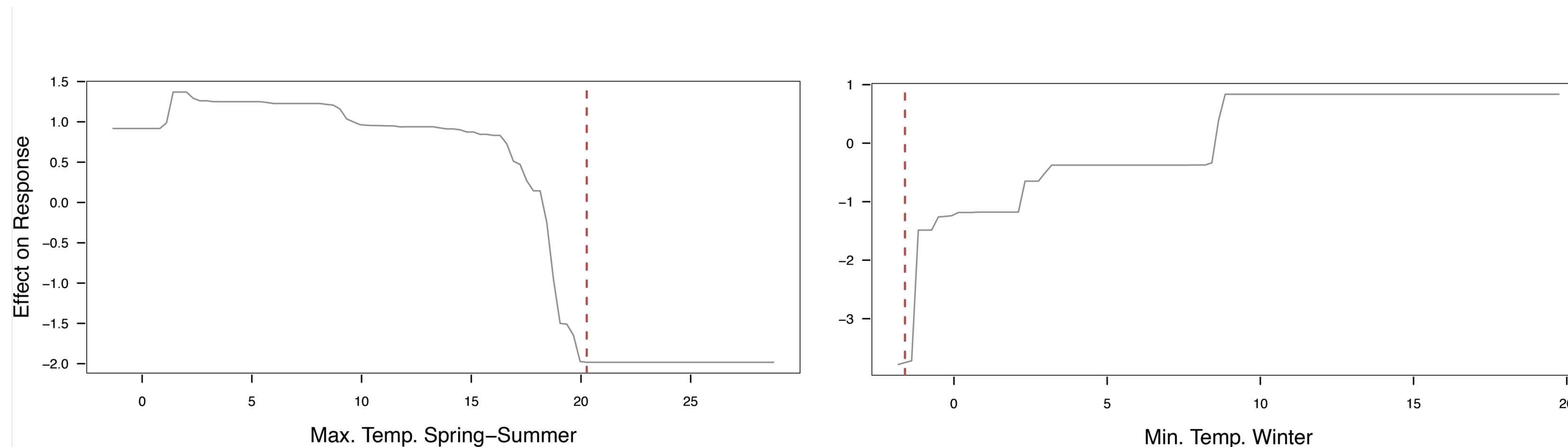
Do predictions align with known species biology, and habitat preferences?

Are the most important predictor variables ecologically relevant?

# Partial dependency plots

Show the relationship between the probability of occurrence and each variable.

The **response is predicted for one environmental variable while the other environmental variables are held constant at their mean**. The x-axis represents the range of values of the environmental variable, and the y-axis gives the probability of occurrence on a scale from 0 (low probability) to 1 (high probability).



**Does the model fit the expectations of ecological theory?**

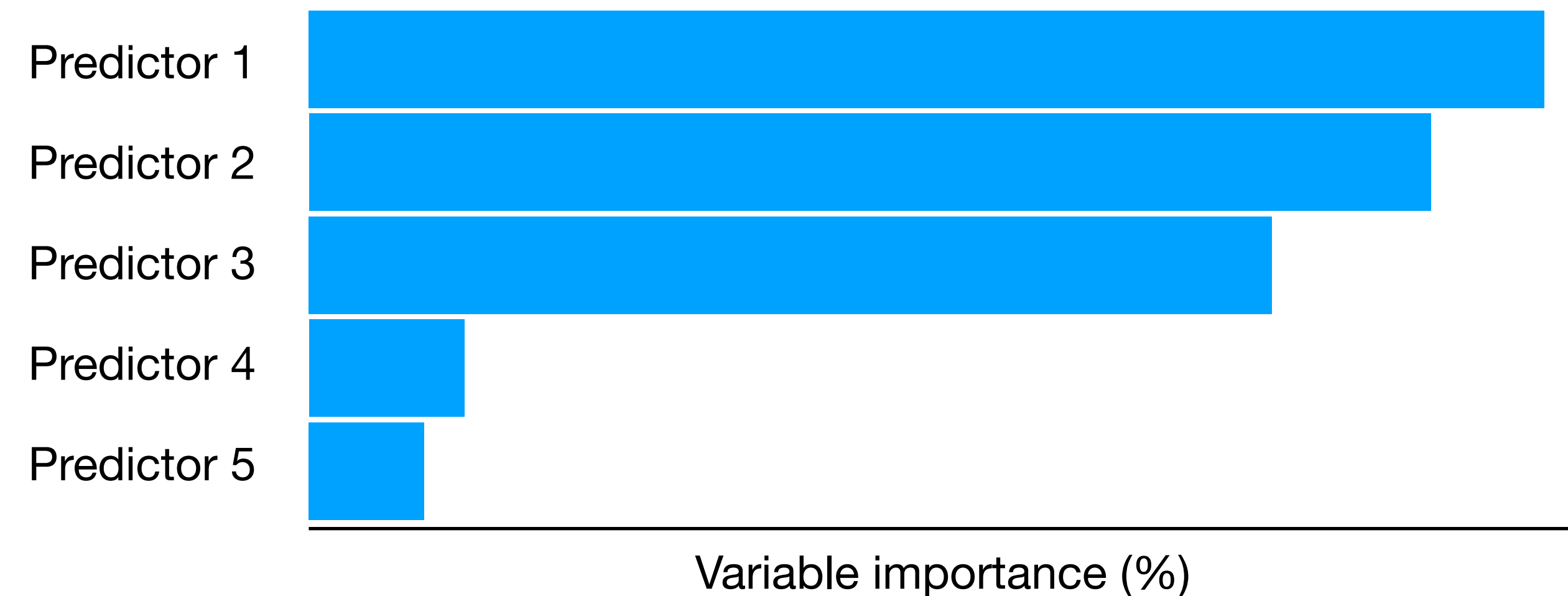
Curve and limiting points match physiological data (reliable model!)



## Relative variable importance

Assists in understanding the contribution of each variable to the model.

The approach is to fit models with and without each variable, in order to determine the potential increase in performance. Without an important variable, a model should reduce its performance.



**Does the model fit the expectations of ecological theory?**